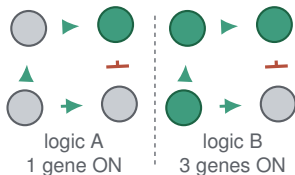


**Logic over wiring:** the model reads local Boolean logic, not just the wiring.

## TRAIN DATA

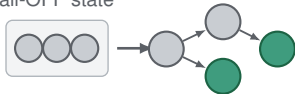
Synthetic Boolean nets ( $n \leq 16$ )

**Same wiring, different logic**  
→ **different reachability**



### Exact oracle

all-OFF state

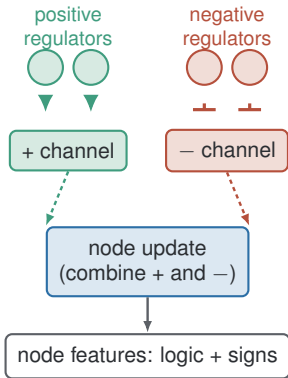


→ per-node reachability labels

→ activation    —| inhibition

## MODEL

Sign-aware, size-agnostic GNN

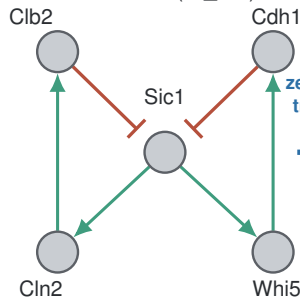


trained on synthetic logic only

● reachable / ON

## TRANSFER

Real GRNs ( $n \leq 70$ )

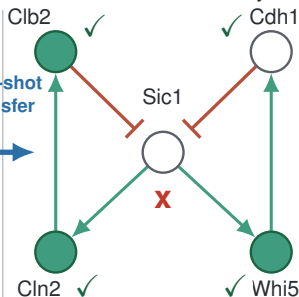


zero-shot  
transfer

**Exact check**  
symbolic BDD, validation only

## OUTPUT

Predicted activatability



**Coloured by prediction:**

- activatable
- not activatable
- ✓ matches exact oracle
- X differs from oracle

**AUROC 0.91 [0.88–0.93]**